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Formal Verification and Web-Based Assessment for ISO/IEC 29110 Using Colored Petri Nets

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***Abstract*—This paper presents a formal verification approach combined with a web-based assessment framework to support ISO/IEC 29110 Project Management (PM) process compliance in Very Small Entities (VSEs). Using Colored Petri Nets (CPNs), the proposed model formally defines and simulates PM-specific processes—namely project planning, execution, control, and closure—verifying their critical properties. A web-based application has been developed to facilitate data collection, trigger simulation, and report evaluation results. Assessment data are systematically transformed into tokens, evaluated through the CPN model, and mapped to compliance outcomes based on ISO/IEC 29110 PM processes. Experimental simulations confirm the accuracy, traceability, and operational efficiency of the proposed system. This integration of formal modeling with web-based automation offers a scalable and reliable solution, combining rigorous verification, systematic process assessment, and practical deployment through an intuitive, user-friendly web interface tailored for small software organizations.**

Keywords—ISO/IEC 29110, Project Management, Colored Petri Nets, Formal Verification, Web-Based Assessment, Very Small Entities.

I. Introduction

The growing emphasis on software process quality and standardization has led many Very Small Entities (VSEs) to adopt ISO/IEC 29110, a lightweight international standard tailored for small organizations. This standard outlines key processes most notably Project Management (PM) and Software Implementation (SI) to support consistent software development and delivery. However, VSEs often face resource and capability constraints that limit their ability to implement and evaluate these processes effectively.

Among the core components of ISO/IEC 29110, the Project Management (PM) process is critical for planning, monitoring, control, and closure of software projects. Ensuring compliance in these areas requires systematic evaluation, clear documentation, and verifiable execution—all of which can be challenging without structured models or automated tools.

This paper addresses these challenges by proposing a formal model of the PM process using Colored Petri Nets (CPNs), which are particularly suitable for modeling concurrent and data-dependent workflows typical in ISO/IEC 29110. Compared to other modeling techniques, CPNs provide explicit token flow, conditional logic, and formal semantics, offering a precise and analyzable representation of task dependencies. To ensure correctness, the model is further verified for key behavioral properties, such as deadlock-freeness, liveness, and reachability. To bridge the gap between theoretical modeling and practical application, the proposed approach is implemented in a web-based assessment system. This system allows users to input evaluation data, simulate CPN models, and obtain compliance results automatically. The automation ensures repeatability, traceability, and ease of use for organizations seeking ISO/IEC 29110 PM process evaluation. The key contributions of this paper are:

- a formal CPN-based model for ISO/IEC 29110 PM processes,
- a critical properties verification to ensure process correctness, and
- an automated web-based system that supports real-time assessment and reporting.

This paper focuses solely on the PM process, excluding Software Implementation (SI).

II. Related Work

The ISO/IEC 29110 standard was introduced to enhance software process capability in Very Small Entities (VSEs), typically organizations with fewer than 25 employees. It provides a lightweight framework focused on Project Management (PM) and Software Implementation (SI). O'Connor and Laporte [4] highlighted the evolution of this standard and its practicality for small software teams.

Colored Petri Nets (CPNs), a formalism combining graphical notation with mathematical semantics, have been widely applied in process modeling and verification [6], [7]. Their strength lies in representing concurrency, state transitions, and conditional behavior. Jensen and Kristensen [6], [7] provide foundational frameworks that support simulation and analysis of complex workflows using CPNs.

Petri Net-based verification has also been used in general process modeling. For example, van der Aalst [10], [11] applied Petri Nets to workflow and process mining. Dechsupa et al. [3] demonstrated hierarchical verification of BPMN models, while

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Thomas et al. [5] extended model-checking to ecological systems, showing the flexibility of Petri Net formalism.

Despite these applications, prior work has not addressed ISO/IEC 29110 PM process modeling specifically. Rodger et al. [2] explored Petri Nets for decision-making in government workflows, but without explicit alignment to ISO/IEC standards. To date, no study has proposed a comprehensive framework integrating CPNs, CTL verification, and automated web-based assessment for ISO/IEC 29110 PM processes.

This paper addresses this gap by offering a formal and automated evaluation system tailored to ISO/IEC 29110, supporting traceable, repeatable assessments for VSEs.

III. Methodology

The methodology consists of four main components: the transformation of ISO/IEC 29110 evaluation criteria into formal tokens, the construction of a Colored Petri Nets (CPNs) model to represent Project Management (PM) workflows, its model verification, and the implementation of a web-based system to automate the assessment process.

A. Evaluation Criteria and Token Transformation

ISO/IEC 29110 defines key Project Management (PM) activities along with corresponding input/output artifacts and evaluation criteria. To enable formal analysis, these qualitative criteria are converted into token-based representations within the CPN model. Each evaluation result, such as the status of a work product or task, is transformed into a Boolean value (true or false) and then mapped to a numeric score:

- true → 1.0 (Satisfied)
- false → 0.0 (Unsatisfied)

Tokens are replaced in the CPN model as 1'(true) or 1'(false) based on each Boolean result, linking assessment outcomes to simulation behavior. For each subprocess, token values from all subtasks are summed to compute the total token value, as shown in (1):

$$\sum Token = \sum_{i=1}^{n} Token_i \tag{1}$$

The average token value is then calculated to derive the process score, as shown in (2):

$$PM_i = \frac{\sum Token}{n} \tag{2}$$

Finally, the overall performance is determined by aggregating the scores of all PM activities, as expressed in (3):

$$\text{AveragePM} = \frac{\sum PM_i}{k} \tag{3}$$

where n is the number of subtasks and k is the number of activities are evaluated. This quantitative transformation provides a consistent and objective basis for evaluating ISO/IEC 29110 compliance, supporting traceable and repeatable assessments. The evaluation criteria follow the rating profile defined in ISO/IEC 29110, where individual task scores are assigned based on percentage completion:

- 0.3 (orange): Partially achieved (15–50%)
- 0.7 (yellow): Largely achieved (50–90%)
- 1.0 (green): Completely achieved (>90%)

The overall process result is marked as "Satisfied" (S) with a green indicator only when all subtasks within the evaluated activities are rated either "L" or "C" and no task is marked "N" (red). Otherwise, the result is deemed "Unsatisfied" (U) with a red indicator. This approach provides clear visual cues and allows stakeholders to quickly interpret assessment outcomes based on color-coded scoring.

B. Formalization of PM Processes Using CPN

The evaluation model is constructed using Colored Petri Nets (CPNs), a formal modeling language that integrates graphical notation with rigorous mathematical foundations. CPNs are particularly well-suited for modeling concurrent, event-driven processes such as software project workflows. Each PM activity is formalized using the following CPN elements:

- Places represent the state of project tasks or deliverables, such as work product updates or project status.
- Transitions capture evaluation logic and task execution conditions.
- Tokens represent the evaluation status of each task, transformed from user assessment input.
- Subpages define modular process groups with structured input and output channels to support hierarchical modeling.

Actions defined within transitions consist of input variables, output results, and expressions for computing numerical scores from Boolean task inputs.

To enhance modularity and enable scalable validation, subpages are created for each PM process group (e.g., PM2: Project Plan Execution). The PM2 subpage receives tokens from PM1 (Project Planning) through a defined subpage input. Evaluation is carried out through six internal transitions (PM2.1 to PM2.6), which apply logic based on user assessment. Progress Status Records are also used in both PM2 and PM3 to reflect task progress. Each evaluation transition in PM2 applies an action expression that processes input tokens and generates an output score. The resulting token is passed back through the subpage output for integration into the top-level model.

The computed PM2 score is then aggregated with the results from PM1 (Project Planning), PM3 (Project Assessment and Control), and PM4 (Project Closure) to determine the overall Project Management capability score. This structure ensures traceable and repeatable evaluation aligned with ISO/IEC 29110.

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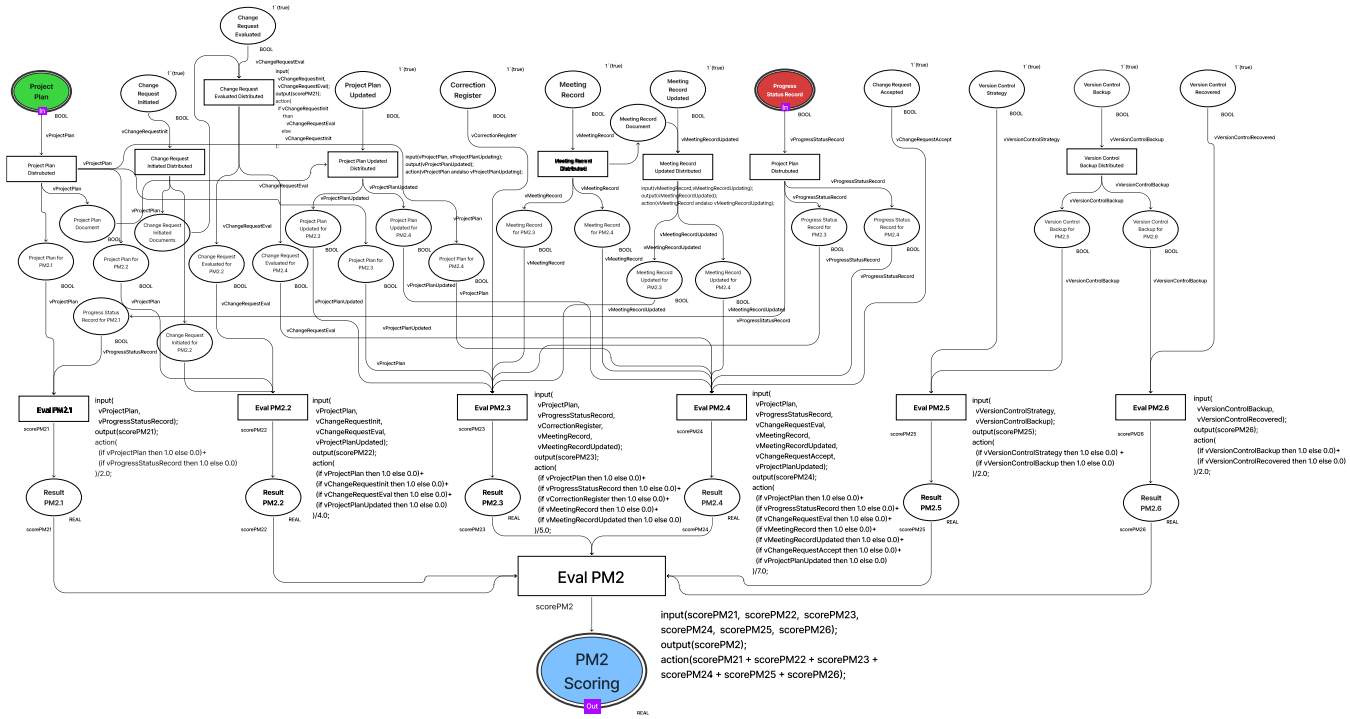


Fig. 1 CPN model for evaluating PM2 in ISO/IEC 29110. CPN subpage model for PM2: Project Plan Execution. The subpage includes input channels (in) receiving tokens from the PM1 evaluation (Project Plan), highlighted in green, and tokens from the PM3 evaluation (Progress Status Record), highlighted in red. Internal transitions (EvalPM2.1 to EvalPM2.6) represent task-specific evaluations, while output channels (out) deliver aggregated PM2 scores to the top-level model, highlighted in blue. The structure reflects token flow, status tracking via Progress Status Records, and formal compliance scoring for all PM2 subtasks

C. Critical Properties Verification

To ensure the correctness and conformance of the proposed formal model, comprehensive state space analysis was performed using CPN Tools across all Project Management (PM) processes defined in ISO/IEC 29110, namely PM1 (Project Planning), PM2 (Project Plan Execution), PM3 (Project Assessment and Control), and PM4 (Project Closure).

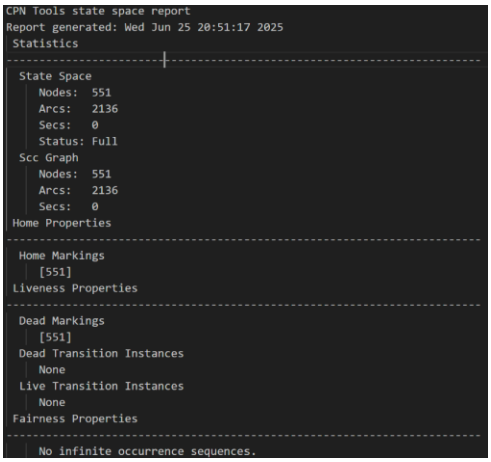
The formal model was iteratively refined and exhaustively verified to satisfy the following critical properties:

- Deadlock-freeness: The system progresses from the initial marking to the final marking without getting stuck, guaranteeing process completion and score generation.
- Reachability: All modeled states and process conditions are accessible, ensuring comprehensive coverage of possible system behaviors.
- Termination Property: Each activity concludes with a well-defined terminating marking, reflecting correct process completion and result reporting.
- Liveness: No transitions or tasks remain permanently inactive unless intentionally designed for process closure.

All activities (PM1–PM4) were successfully verified to satisfy these properties, confirming system-wide correctness, completeness, and traceability.

As a representative example, the state space analysis for PM2: Project Plan Execution is presented. The generated state space consists of 551 reachable markings and 2,136 arcs, as shown in Fig. 2, which displays the state space report generated by CPN Tools. The log confirms that all markings are classified as dead markings, meaning the process terminates as designed upon task completion and evaluation. No infinite occurrence sequences or dead transitions were detected, validating deadlock-freeness, finite behavior, and conformance to ISO/IEC 29110 requirements.

This representative result demonstrates the rigor of the formal modeling approach, with full-system verification completed for all PM activities.



```
CPN Tools state space report
Report generated: Wed Jun 25 20:51:17 2025
 Statistics
------------------------┼-----------------------------------------
  State Space
     Nodes:  551
     Arcs:   2136
     Secs:   0
     Status: Full
  Scc Graph
     Nodes:  551
     Arcs:   2136
     Secs:   0
 Home Properties
-----------------------------------------------------------------
  Home Markings
     [551]
 Liveness Properties
-----------------------------------------------------------------
  Dead Markings
     [551]
  Dead Transition Instances
     None
  Live Transition Instances
     None
 Fairness Properties
-----------------------------------------------------------------
     No infinite occurrence sequences.
```

Fig. 2 CPN Tools state space analysis log for PM2 process, showing 551 reachable markings, 2,136 arcs, and confirmation of process termination and deadlock-freeness.

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D. Integration with Web-Based Assessment System

The formal evaluation model is embedded within a web-based assessment system to enable practical application in Very Small Entities (VSEs). The backend component, implemented using Python and Robot Framework, handles token injection, simulation triggering via CPN IDE, and result parsing. Each PM activity's compliance score is computed through token-based aggregation logic and mapped to standard ISO/IEC 29110 rating categories.

To accommodate non-expert users, the interface abstracts the underlying formal model and verification logic. This enables self-assessment without requiring prior knowledge of CPN, while still preserving traceability and rigor in evaluation. The assessment results are compiled into a structured Process Capability Rating Chart, which presents performance across all PM activities, the number of evaluated tasks, total scores, and the degree of achievement based on predefined color-coded classifications. A typical interaction flow is illustrated in Fig. 3, showing how user input triggers simulation, result processing, and report generation.

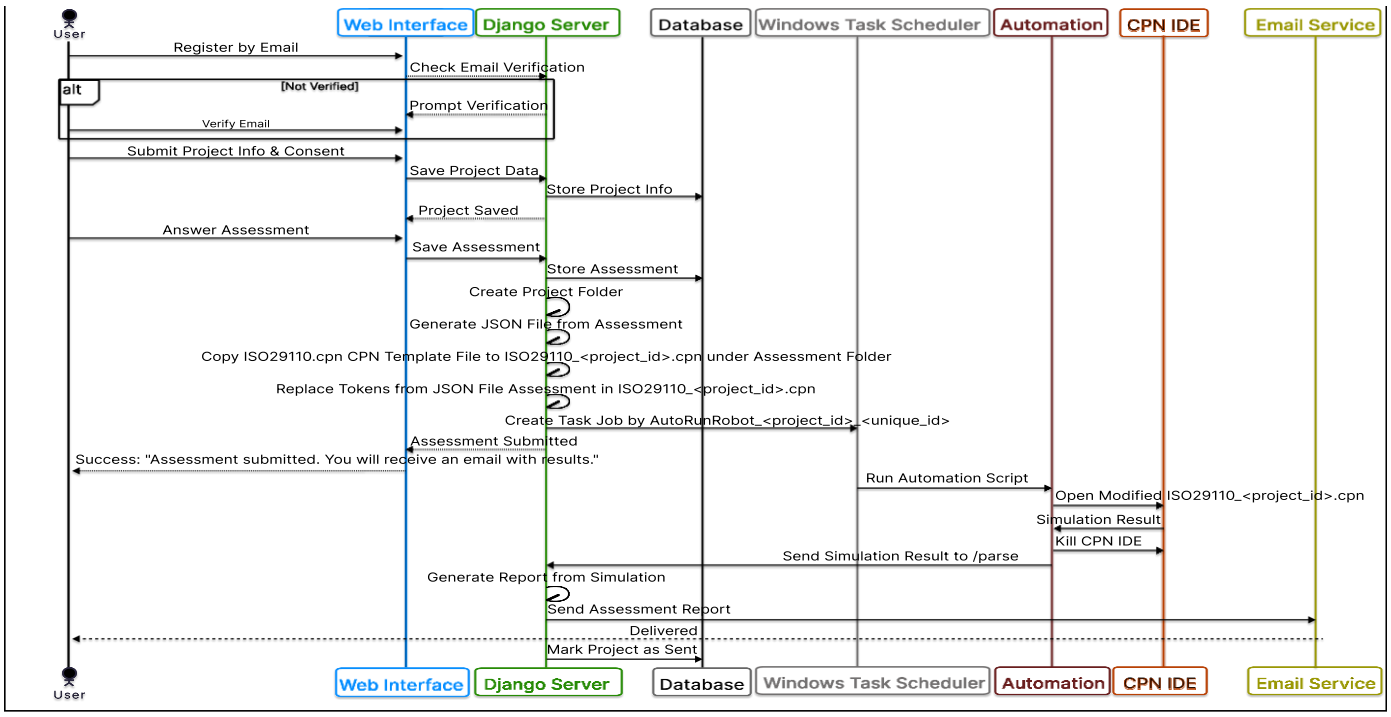

Fig. 3 Sequence Diagram of the ISO/IEC 29110 Web-Based Assessment System Integrating CPN IDE and Email Report Delivery

In addition to displaying results on screen, the system generates structured PDF reports that are automatically sent via email to users for documentation, review, or external audit purposes. The report includes a Process Capability Rating Chart that summarizes the degree of achievement across PM areas, number of evaluated tasks, and compliance levels using color-coded categories. To enhance usability, a web-based interface enables project members to submit binary responses (Yes/No) aligned with PM activities. An example screen is shown in Fig. 4. The system, implemented with Django, Python, and Windows Task Scheduler, supports automated simulation via CPN IDE.

The complete CPN model integrated into the system is shown in Fig. 5. It demonstrates the hierarchical structure, modular evaluation subpages, and token-based assessment mechanisms that underpin the web-based system's formal verification capabilities.

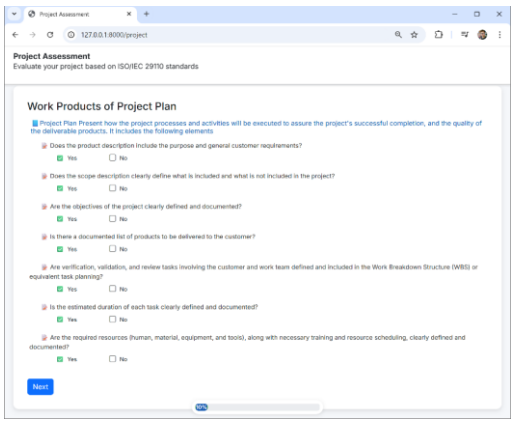

Fig. 4 Web-based self-assessment interface for PM1(Project Planning), allowing users to input evaluation results aligned with ISO/IEC 29110.

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To enhance usability, the system automatically generates PDF reports summarizing assessment results, which are emailed to users for formal documentation, review, or external audit purposes.

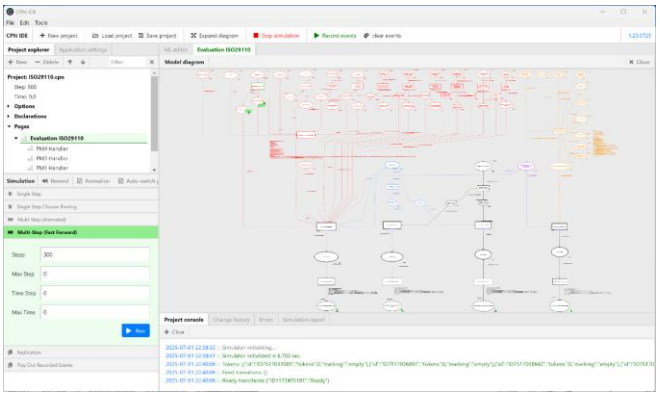

Fig. 5 Complete CPN model developed using CPN IDE, illustrating hierarchical structure, evaluation subpages, and token-based assessment logic

IV. Experimental Results and Evaluation

To demonstrate the applicability of the proposed formal evaluation model, a simulated assessment scenario was created using the developed web-based system. The system is designed to support ISO/IEC 29110 Project Management (PM) processes, enabling self-assessment and formal verification through token-based Colored Petri Net (CPNs) models. For demonstration purposes, a test dataset was submitted through the web interface. The system transformed user inputs into token representations, performed evaluation logic via CPN IDE, and generated a complete assessment report. The evaluation covered four PM activities: PM1 (Project Planning), PM2 (Project Plan Execution), PM3 (Project Assessment and Control), and PM4 (Project Closure). Fig. 6 presents an example of a failed assessment automatically generated by the system. The chart illustrates the scoring results, degree of achievement by activity, quality performance, and traceable evaluation links. In this case, several PM tasks were marked as "Not achieved" (N) or "Partially achieved" (P), leading to an overall score of just 62.25% and a final rating of "U" (Unacceptable). The individual task scores of 0.3 (Partially achieved), 0.7 (Largely achieved), and 1.0 (Completely achieved) are averaged and scaled to generate the final percentage-based assessment result shown in Fig. 6. This demonstrates the system's ability to process token-based evaluations, highlight weaknesses, and generate structured outputs according to ISO/IEC 29110 guidelines.

This demonstrates the system's effectiveness in identifying underperformance and guiding Very Small Entities (VSEs) toward targeted process improvements. These experimental results validate the system's ability to perform accurate, automated, and traceable assessments across multiple PM activities. Future enhancements include full-scale deployment to a production server and potential extension to all ISO/IEC 29110 process. Currently, the system has been implemented and tested on a local Windows 11 environment. Full deployment to a production server will be considered in future work.

Process Capability Rating Chart

Project Description:
Project Name: **RFR Migration and Readiness Project**
Project Period: **1 เมษายน 2024 ~ 28 กุมภาพันธ์ 2025**
Assessment Date : **24 มิถุนายน 2025**

Process Capability Rating Chart																					
พื้นที่กระบวนการ (Process Area)	ระดับความสำเร็จ (Degree of Achievement)	คะแนนทั้งหมด (Total Score)	ประสิทธิภาพเชิงคุณภาพ (Quality Performance)	กิจกรรมที่ดำเนิน (Task Lists)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
PM1	N	9.42	0.63	15	C	C	C	P	C	C	L	P	C	N	P	P	P	N	N
PM2	N	4.06	0.68	6	N	L	P	L	C	C									
PM3	N	1.00	0.33	3	N	P	P												
PM4	L	1.70	0.85	2	C	L													
Total PM Score		16.18	2.49	26	N = 5														
				62.25%															

Process capability level ratings

Final Findings	Degree of Achievement	Description
PM	U	FAIL

The degree of achievement	Color score
N: Not achieved	
P: Partially achieved	
L: Largely achieved	
C: Completely achieved	
NA: Not applicable	

Fig. 6 Example of a failed assessment result. The chart illustrates low achievement levels across key PM areas, resulting in an overall unacceptable (U) rating, effectively highlighting areas needing improvement.

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V. Discussion and Contributions

This section highlights the strengths, weaknesses, and broader implications of the proposed system, incorporating comparative analysis with traditional approaches and performance evaluation under concurrent usage.

A. *Strengths of the Proposed System*

The system demonstrates several advantages over conventional self-assessment methods (e.g., Excel- or Word-based documentation). As summarized in Table 1, the developed system provides:

- Reduced Evaluation Effort: Assessment time depends on the number of questions.
- Improved Accuracy: Token-based verification ensures logical consistency, minimizing subjective interpretation errors.
- Enhanced Traceability: Unlike traditional approaches that lack backtracking capability, the proposed system supports systematic verification and explicit activity-to-process mapping.
- Lower Human Error Risk: Automation significantly reduces the likelihood of assessor misinterpretation.

These findings reinforce the system's robustness, combining the formal rigor of Colored Petri Nets (CPNs) with a practical, web-based interface suitable for Very Small Entities (VSEs).

B. *Weaknesses and Limitations*

Despite its contributions, several weaknesses remain. The experimental results indicate performance constraints when the number of concurrent users increases (Fig. 7). While the system performs efficiently with small to medium workloads, the average report generation time grows substantially as concurrent users exceed 20, revealing scalability challenges.

In addition:

- Evaluation Environment: Current validation was conducted in a controlled, simulated setting, without extensive real-world testing across diverse organizations.
- Scope Restriction: The implementation is limited to the Project Management process of ISO/IEC 29110, excluding other key processes such as Software Implementation.
- Scalability and Interoperability: Further investigation is required to optimize distributed deployments and extend interoperability with cloud-based platforms or related standards (e.g., ISO/IEC 12207, ISO/IEC 25010).

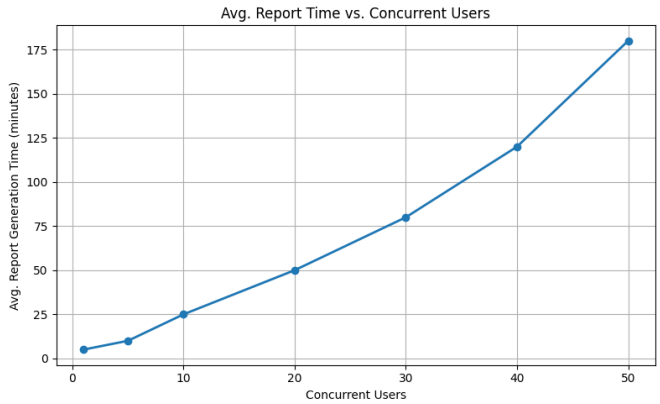

Fig. 7 Average Report Generation Time versus Concurrent Users

Table 1 Comparison between Conventional Self-Assessment and the Developed System

Comparison Criteria	Conventional Approach (Excel, Word)	Developed System (Web Application, CPN IDE)
Tools Used	Excel, Word	Web Application, CPN IDE
Assessment Time	Depends on the number of questions	Depends on the number of questions
Accuracy in Verification	Low – relies on human's interpretation	High – token-based automated verification
Risk of Human Error	High	Low
Automated Summary Capability	None	Available – automated reporting system

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C. Future Work

Future efforts should address the identified limitations by:

- Conducting empirical validation with industry VSEs to enhance credibility.
- Expanding coverage to additional ISO/IEC 29110 process, particularly Software Implementation.
- Improving scalability through distributed or cloud-based deployment, ensuring robustness under heavy concurrent usage.

Overall, the proposed system combines methodological rigor with practical feasibility, outperforming traditional approaches in efficiency, traceability, and accuracy, while also highlighting challenges related to scalability and scope. These contributions mark an important step toward making ISO/IEC 29110 self-assessment more reliable and accessible for very small entities.

VI. Conclusion

This paper introduced a formal evaluation model to assess ISO/IEC 29110 Project Management (PM) process compliance by employing Colored Petri Nets (CPNs). The model formalized PM tasks as token-based workflows, enabling structured simulation and automated compliance verification. Integrated into a web-based self-assessment platform, the approach allowed users to input evaluation data, automatically transform it into tokens, and obtain structured reports summarizing compliance results.

The experimental evaluation demonstrated that the system achieves three key outcomes:

(1) enhanced accuracy, achieved through token-based formal verification that ensures logical consistency.

(2) improved traceability, provided by explicit mapping of tasks to ISO/IEC 29110 processes, enabling clear compliance tracking; and

(3) greater efficiency, realized by reducing assessment time and minimizing human error compared with conventional methods. The generated reports, including process-level scores, task-level classifications, and color-coded indicators, further support both internal reviews and external audits.

In summary, the proposed framework successfully bridges formal verification and practical assessment, providing Very Small Entities (VSEs) with a reliable, repeatable, and user-friendly solution for ISO/IEC 29110 compliance. By combining the rigor of CPNs with the accessibility of a web-based system, the work contributes to advancing process discipline and fostering broader adoption of international standards among small software organizations.

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